

Comprehensive Survey on Load Balancing Techniques in Cloud Computing Environments

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ABSTRACT

The notion of load balancing is necessary to maximise the use of cloud resources due to the daily increase in internet traffic. For the system's performance and stability, load balancing becomes essential. Consequently, an algorithmic strategy is needed to balance work among virtual machines (VMs) and improve system performance. Techniques: To provide load balancing and QoS, task scheduling methods are employed. Cloud computing is an entirely internet-based method in which all information and programs are stored on a cloud, which is made up of thousands of machines connected in a complex way. By guaranteeing optimal use of available resources, a load balancing algorithmic strategy seeks to improve the response time of user-submitted applications. The primary goal of load balancing techniques is to accelerate the execution of programs on resources whose workload fluctuates in unanticipated ways during runtime.

Keywords: Cloud load balancing, Round robin algorithm, Load Balancing, Makespan Response Time, Task Scheduling.

I. INTRODUCTION

Cloud computing is a completely internet-based method in which all information and programs are stored on a cloud, which is made up of thousands of intricately connected computers. In order to supply computers or other devices with shared resources—hardware, software, and information—on demand, cloud computing integrates the ideas of distributed

and parallel computing. as communication and information technology have advanced. As the need for computing rises, distributed computing systems are becoming more and more prevalent. A large scale distributed systems are required with a large number of servers. Appropriate work distribution to every node is essential for the effective operation of distributed systems; if tasks are assigned at random, some nodes may gets overburdened while others get

inactive; this is prevented by employing content-based distributed clustering with random walk search for effective dynamic load balancing.

Clusters are appealing platforms for deploying applications at scale for outstanding performance, while distributed networks provide large-scale sharing of resources and system integration. In the clustering environment, balance of load is necessary for the appropriate distribution of user demands.

A technique for dividing up the burden among several servers, network ports, hard discs, or other computer resources is load balancing. Large, powerful (and costly) computing hardware and network infrastructure are the foundation of typical datacenter implementations. These components are vulnerable to the same risks as any other physical device, such as failures in hardware, power and/or network outages, and resource constraints during periods of high demand..

Nowadays, cloud computing is used in practically every aspect of our lives. However, despite meeting countless user demands, it has a few obstacles to overcome. It asks us to concentrate on problems like security, authentication, fault tolerance, load balancing, and availability in addition to offering us features like virtualisation, resource sharing, ubiquity, and utility computing.

One of the finest technologies of the decade is cloud computing. due to its simple and adaptable architecture, a lot of businesses are looking to develop and deploy clouds. As a result, more and more people are using the cloud. Even if there are public, private, and hybrid cloud models, these clouds may still have dependability problems. Cloud models make use of virtualisation technology, which makes it easier to divide a single data centre (DC) or high-power server into several units..

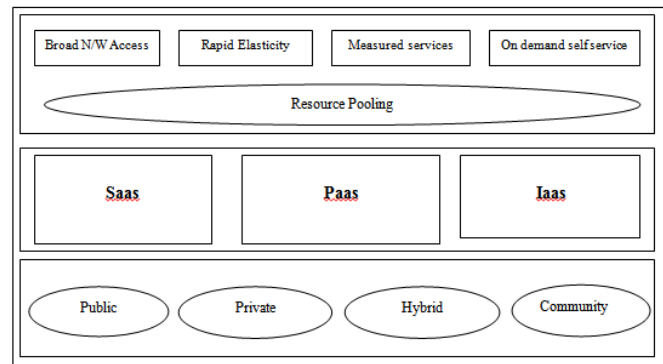


Fig.1 Cloud Computing Architecture

II. LITERATURE SURVEY

Tariq A. Murshedi et al. [2024] "In order to build robust and ecologically conscious engineering systems, this study investigates the relationship between sustainability, load balancing, request management, resource variety, system capabilities, and cost-effectiveness. The goal of lasting peace is included in sustainability, which is a fundamental component of social advancement.

on Earth, including social welfare, economic growth, and environmental protection. The prudent and effective use of resources is closely linked to technological progress. An essential part of computer communication, load balancing aims to distribute workloads to computing resources as efficiently as possible in order to reduce bottlenecks and redundancies and guarantee seamless network functioning. The quantity and kind of requests become crucial factors in this paradigm, dictating how well resources can be allocated to handle the incoming tasks [01].

Ellakkiya Mathanraj et al. [2024] With the help of the promising technology known as cloud computing, customers may meet their computing needs as they arise. This study introduces a new load balancing PCGRLB is a method for allocating several jobs in the shortest amount of time. Using the contingency correlative principal component projection, the PCGRLB first analyses the virtual machine resources. Consequently, the virtual machine's balanced, less burdened, and overloaded loads are determined. The

PCGRLB approach then uses gradient weighted MapReduce round-robin load balancing to distribute the workload among the cloud server's virtual machines. The load balancer migrates tasks at the lowest possible cost. Lastly, the reduction phase aggregates the virtual machine's output and offers the results of the final product. The performance analysis results demonstrate that the suggested algorithm successfully achieves more effective load balancing accuracy with maximum flow and less makespan, emigration cost, and response time than the conventional methods after an experimental assessment of the PCGRLB technique and existing methods.

Yogesh Lohumi et al. [2023] Cloud Computing (CC) provides on-demand IT services and resources over Internet network, including pay-for-you-go options. The CC paradigm offers several services, including computational services, storage spaces, deployment situations, and infrastructure-related resources. The goal The purpose of this study is to examine load unbalancing, a multi-variant, multi-constraint problem that significantly impairs the system's scalability, efficacy, and performance in cloud computing. Cloud service providers continue to aim for server workload distributed equilibrium. Load balancing techniques use overloading and underloading to address the unbalancing problem. In order to improve the quality of service, this research article presents a thorough structural literature analysis of load balancing and its component domains, taking into account a number of characteristics including scalability, make-span, and throughput. A comprehensive and well-structured taxonomy of all load balancing algorithms

Techniques, functionality, and types are also offered according to the nature of the system state. The survey's primary focus is on load-balancing algorithms that are static, dynamic, hybrid, and inspired by nature [03].

Mana Saleh Al Reshan et al. [2023] The dynamic provisioning of resources to deliver services to end

users via the internet is known as cloud computing. A number of issues, including resource discovery, security, scheduling, and load balancing, must be resolved for cloud computing to become a reality. The most important of these research concerns is load balancing difficult one. As a result, research into different static and dynamic algorithms to attain optimal outcomes has become more significant in recent years. Swarm Intelligence (SI) is suggested in this study as a cloud computing loadbalancing solution. The load balancing convergence time with global optimisation is not taken into consideration by any of the several options that have been studied in the literature, including genetic algorithms, ACO, PSO, BAT, GWO, and many more. Among these methods, this research focusses Particle Swarm Optimization (PSO) and Grey Wolf Optimization (GWO). This work proposes a combined GWO-PSO strategy that leverages the advantages of global optimisation and quick convergence. Together, these two strategies improve resource allocation and system efficiency, resolving the load-balancing issue [04].

Abderraziq Semmoud et al. [2022] "In addition to offering several benefits over on-premise IT infrastructures in terms of cost, flexibility, and availability, clouds give customers on-demand access to massive processing and storage capabilities. But there are still a lot of issues with this new paradigm, and in this article we tackle the load balancing issue. Despite the fact that several methods have been put forth to distribute the load across the servers, the majority of them result in unstable systems and are excessively susceptible to variations in the cloud's demand. In this work, we provide a novel adaptive starvation threshold-based distributed load balancing method. The suggested method's performance was contrasted with that of a popular load balancing algorithm that was modelled after the behaviour of honey bees (HBB). According to the testing findings, using the suggested load balancing method provides notable improvements in performance and a notable

decrease in the quantity of migrations in contrast to the HBB algorithm's performance [05].

Saurabh Gupta et al. [2019] One of the biggest problems with cloud computing is load balancing. We covered a number of static load balancing approaches in this study, including algorithms for dynamic load balancing. Additionally, we spoke about the traits, benefits, and drawbacks of several load balancing algorithm types [06].

Dr Syeda Gauhar Fatima et al. [2019] Cloud computing is a type of utility software that may transform software companies and increase the program's appeal. The way the IT industry used to build hardware and software has changed as a result of cloud computing. The increase in website traffic and various. As the number of services grows daily, load balancing becomes a major area of study. Cloud computing is a system that hosts, stores, and networks the various computer components using virtual machines rather than physical ones. As an alternative to actual computers, load balancers are used to distribute loads among many virtual machines so that no machine is overloaded or underloaded. Due to the possibility of node failure, load balancing must be done correctly[07].

Aida A. Nasr et al.[2019] "In a cloud computing context, a novel two-phase SAAC method for cloudlet scheduling is put forward. The suggested SAAC method can

At low time complexity, schedule a large number of cloudlets onto the available virtual machines (VMs) while taking into account several resource restrictions. It improves system performance overall and maintains the virtual machines in a high degree of balance.

The novel SAAC technique outperforms the SA and ACO algorithms in terms of efficiency to compare them [08].

III.METHOD

3.1. Load Balancing

Spreading the incoming request load among the available request execution nodes is the simplest way to implement load balancing in your application. The purpose of load balancing is to prevent the executing nodes from becoming overburdened with user requests. The performance is significantly improved by the cloud manager's small action. There are essentially two kinds of load balancing algorithms based on how they are implemented, according to.

3.1.1. Static Load Balancing Algorithm:

It is independent of the system's present condition. It determines which host the request will be performed at before it is set up..

3.1.2. Dynamic Load Balancing Algorithm:

Load balancing decisions are based on the system's present condition. The load balancer performs requests at the appropriate host after analysing the load statistics at each available host. We can implement the load balancing notion at a few levels in the existing cloud architecture. Load balancing may be applied to the Virtual Machine Monitor (VMM), Server cluster, and Service Broker. A simple cloud architecture, as depicted in [10], is presented in Fig. 1. Region N shows the users' geographic location. A user base is a collection of users who send requests. The appropriate server that can complete user requests will be found with the aid of Service Broker. Here, we've demonstrated a virtual machine load balancer at the cluster of running nodes. In more dire circumstances, the VM load balancer decides to migrate the virtual machine instead of using VMM to load the virtual machine [1]. The load balancer distributes a virtual machine (VM) among all accessible nodes in a cyclic fashion to the requesting node in the basic round robin method [3]. The primary benefit of this approach is that it ensures fairness by allocating an equal number of virtual machines (VMs) to each node and using all resources in a balanced sequence.

3.1.3. Round Robin

A straightforward technique for delivering basic fault tolerance or load balancing servers. Several similar servers are set up to deliver the same services. Although each has a distinct IP address, they are all set up to utilise the same Internet domain name. All of the distinct IP addresses connected to the Internet domains are listed on a DNS server. Upon receiving requests for the IP address linked to the Internet domain name, the addresses are provided in a consecutive rotation.

3.1.4. Weighted Round Robin

This improves upon the straightforward Round Robin load balancing technique. Every server in the pool is assigned a static numerical weighting in the weighted version. Requests are routed to servers with higher ratings more often.

3.1.5. Least Connection

When allocating requests, neither Weighted Round Robin nor Round Robin take the current server load into account. The current server load is taken into account by the Least Connection technique. The server handling the fewest active sessions at the moment receives the current request.

3.1.6. Weighted Least Connection

Extends the Least Connection approach. Each server is assigned a numerical value, same like in the Weighted Round Robin technique. This is how the load balancer assigns responses to servers. The server with the greater weighting will get the new request if two servers have the same number of active connections..

3.1.7. Weighted Response Time

This technique identifies the server that is responding the quickest at a given moment by using the response data from a server health check. That server then receives the subsequent server access request. This makes sure that no new requests are delivered to services that are overloaded and will react more slowly. This enables the demand to gradually balance out throughout the available server pool.

3.1.8. Software Defined Networking (SDN) Adaptive

SDN Adaptive integrates information about the network's condition at lower layers with an understanding of upper networking layers. When determining how to distribute requests, knowledge of the network from Layers 2 and 3 is mixed with information about data from Layers 4 and 7 of the network. This enables data on the quality of the network infrastructure, the state of the servers and the applications that run on them, as well as the degree of network congestion, to all be taken into consideration when making decisions about load balancing..

3.2. Cloud Architecture

The four main categories of cloud service models that are displayed by a particular cloud infrastructure are SaaS, PaaS, IaaS, and Data as a Service (DaaS).

3.2.1. Software as a Service (SaaS)

A cloud computing approach called Software as a Service (SaaS) removes the requirement for users to install or update software programs on their local devices by delivering them online. SaaS offers increased flexibility and scalability by enabling both individuals and enterprises to access a vast array of apps via web browsers. Users may only pay for the services they utilise using this subscription-based business model. SaaS suppliers take care of all technical facets, such as infrastructure, security, and upgrades, so that clients may concentrate on their primary responsibilities without having to worry about software upkeep. Project management software, customer relationship management (CRM) tools, and productivity suites like Microsoft 365 and Google Workspace are a few examples of SaaS applications. This method has completely changed the way that software is distributed.

3.2.2. Platform as a Service (PaaS)

Platform as a Service A cloud computing architecture called Platform as a Service (PaaS) gives developers a comprehensive platform to create, launch, and

maintain applications without having to deal with the hassle of maintaining underlying infrastructure. PaaS provides a variety of cloud-hosted tools, services, and environments, such as databases, middleware, development frameworks, and programming languages. With this architecture, developers can concentrate on building and delivering code, allowing the PaaS provider to handle things like security, scalability, and server maintenance. PaaS is a useful tool for contemporary software development as it streamlines the development lifecycle and lowers operational expenses, enabling businesses to improve collaboration, speed up application development, and shorten time-to-market..

3.2.3. Infrastructure as a Service (IaaS)

One of the fundamental cloud computing models, Infrastructure as a Service (IaaS), provides virtualised computer resources via the internet. It gives companies on-demand access to crucial infrastructure elements including operating systems, networks, storage, and virtual machines. IaaS allows businesses to pay for just what they use, expand their resources dynamically, and do away with expensive on-premises gear. Businesses may install and manage their applications using this model's flexibility, dependability, and control while still having complete control over the underlying infrastructure. IaaS is especially well-suited for businesses looking to reduce expenses, improve their capacity for disaster recovery, and foster innovation by using scalable and adaptable infrastructure solutions.

3.2.4. Data as a Service (DaaS)

On-demand virtualised storage supply turns into a distinct cloud service for data storage. Observe that DaaS may be considered a unique kind of IaaS. The driving force is the high upfront costs associated with on-premise business database systems, which include dedicated servers, software licenses, post-delivery services, and internal IT maintenance. Instead of paying for the site licence for the full database, DaaS enables users to only pay for the services they use. Some DaaS offerings offer table-style abstractions in

addition to more conventional storage interfaces like file systems and RDBMS. These abstractions are made to scale out to store and retrieve large amounts of data in a short amount of time, which are frequently too big, too costly, or too slow for the majority of commercial RDBMS to handle.



Fig.2 Basic Cloud Architecture

IV. CONCLUSION

The load balancing strategy for cloud computing systems was examined in this research. The fundamentals of cloud computing are covered in this study, along with load balancing research difficulties. This succinct analysis of cloud computing load balancing aims to highlight the most current findings in the area. A few research articles that addressed various facets and methods of load balancing for cloud computing were examined. The architecture of cloud computing is another area of emphasis. One of the numerous difficulties in the cloud computing world is load balancing. Review of the introduction of load balancing techniques in this study.

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